Additional file 1: supplementary methods and sensitivity analyses

GDTCR scientific revision

Corruption and reconstruction

Each single-chain sequence is assigned a masking rate of 15% or 30% with equal probability. Exactly the larger of one and the rounded rate times sequence length is selected without replacement. Each selected residue independently receives a mask token (80%), a uniform canonical amino acid (10%) or its original token (10%). Deletion starts are sampled uniformly, with span lengths uniformly 1-3, until at least the rounded 8% target is covered; spans can overshoot that target. Deletion follows masking and can overlap selected sites. The damaged mask includes selected-but-unchanged residues and deleted sites. The aligned-copy control copies the corrupted aligned token and uses a mask at deleted sites. Reconstruction includes EOS in token NLL; residue accuracy is separately evaluated on original residue positions. No reconstruction metric is interpreted as partner recovery.

Controls and training

The conditional and constant-source generators share initialization, target sampling, optimizer and nominal downstream budget. Constant-source input keeps the chain token and removes source residues. Random initialization removes pretraining and retains downstream settings; total training computation is not matched. Shuffled-input evaluation permutes queries without fixed points and keeps the original recorded-partner labels. A separate copy of the fine-tuned generator initializes the ranker. Its copied encoder is trainable; the source generator is not updated. All paired fits use the same protocol-specific training edges. Seeds, selected checkpoints and actual exposure accompany Additional file 3.

AdamW uses weight decay 0.01 and gradient-norm clipping at 1.0. For denoising and generator training, the per-step learning rate is the nominal rate multiplied by $\min(1, s/w)$ and $[1 + \cos(\pi \min(1, s/S))]/2$, where s is the current step, w the warmup and S the nominal maximum steps. This applies cosine decay over the whole nominal run, including warmup. Ranker training uses its fixed stated learning rate. Training may stop early; actual exposures, not only nominal limits, are retained.

Table 1. Core generator controls. All conditions retain the protocol-specific test query population and fixed primary sampling rule.

Condition	Comparison purpose
Pretrained conditional	Uses the input chain after denoising pretraining.
Constant-source target prior	Removes source residues while matching initialization, target sampling and downstream optimization.
Random initialization	Removes denoising pretraining; the downstream budget is matched, total computation is not.
Shuffled input	Breaks the input-recorded-partner association at evaluation using the fitted conditional model.

Constructed negatives and candidate construction

The primary ranker first uses multi-positive contrastive training, then equally weighted positive and constructed-negative binary cross-entropy. Three negative draws per positive are drawn from supervised-training combinations, excluding known training partners. The mixed policy combines random and length/terminal matching; actual fallbacks and draw counts are recorded. Unrecorded combinations are not biological negatives. Sensitivities change ratios to one or five, or use random-only draws. Fixed retrieval pools inject all role-specific recorded positives and up to 100 unrecorded targets from that role's universe. Other recorded partners are excluded from distractors. Random, length, terminal-motif and normalized-edit matching are deterministic at fixed seed. All scorers use the same frozen lists. Free sampling instead makes 64 draws with temperature one and top-p 0.95; 6-35 canonical residues and observed EOS are required. Invalid and duplicate draws are never replaced. Training-target exclusion is an explicitly separate policy that can delete legitimate exposed positives.

For each negative draw, the source direction is chosen with equal probability. Under the mixed policy, a hard draw is chosen with probability 0.5. Candidate training targets within one residue of the reference target length are sorted by absolute length difference, mismatch count over the first and last three residues, then sequence. The draw is uniform over the first 64 eligible targets after known-training-partner exclusion. If none remain, sampling falls back to uniform random eligible training targets. Sampling is with

replacement; no held-out pairing labels filter training negatives. The two BCE classes each receive one-half of the loss weight at every negative ratio.

Fixed-pool length matching sorts eligible role targets by the minimum length gap to a role-specific positive, then normalized full-CDR3 edit distance and sequence. Motif matching uses the sum of Levenshtein distances over the first and last three residues, then full normalized edit distance and sequence. Edit matching uses full normalized distance and sequence. Random pools sample without replacement. Candidate targets are lexicographically stored, and score ties use that order. Truth-containing retrieval is an explicitly constructed diagnostic; no such positive-directed pool construction is used in free sampling.

Uncertainty and event reporting

Query-macro metrics weight inputs equally, with directions averaged equally for the primary endpoint. The bootstrap uses 2000 resamples of complete test graph components, shared across directions and paired model contrasts. Directional tasks use source-query groups. The repair of unseen-gamma selection retains this directional uncertainty unit. The within-giant-component experiment has one original component and reports no inferential interval. Intervals condition on fitted models and saved samples, without multiplicity adjustment. A zero-event [0,0] interval is not a population upper bound. Across primary splits, seeds are averaged first and the three split means have a descriptive SD; neither fits nor repeated splits are independent biological replicates.

Supplementary sensitivity results

Pretraining mixture and downstream sampling have distinct effects

The two-seed downstream NLL means for square-root, raw and balanced pretraining are, respectively, 1.256/0.754, 1.255/0.757 and 1.259/0.755 for γ→δ/δ→γ on split 17 (Figure S1). Raw-minus-square-root equal-direction differences are approximately 0.001 nats per token in both fits, with paired intervals crossing zero. Balanced-minus-square-root differences are approximately 0.002 in both fits, with intervals above zero. These small NLL differences do not select a new primary mixture after testing.

Restricting pretraining to one chain has larger directional effects. Gamma-only initialization gives downstream NLL 1.294/0.755, while delta-only initialization gives 1.257/0.779. On frozen denoising validation, gamma-only pretraining yields 39.6% selected-site accuracy for delta, and delta-only pretraining yields 27.3% for gamma, compared with 64.4% and 70.8% under the square-root mixture. The results are

consistent with chain-specific reconstruction exposure affecting the target-sequence prior. Equal total sequence draws do not equate chain-specific or token exposure, and these teacher-forced differences do not demonstrate reliable free partner recovery.

Edge-uniform paired training gives NLL 1.263/0.759, compared with 1.256/0.754 under query-uniform training. Its equal-direction excess is approximately 0.006 nats per token in both fitted seeds, with paired intervals above zero. The nominally direction-budget-matched unidirectional models give NLL 1.258 for γ→δ and 0.756 for δ→γ, close to the bidirectional model. All these settings retain sparse free-recovery events. Their actual early-stop and token exposure records accompany the matched nominal budgets; the comparison does not establish a universal benefit from sharing the two directions.

Changing negative training does not resolve candidate scarcity

Random-only and mixed negative training at ratios one, three and five reuse the same split-17 generator candidates and retrieval pools (Figure S2). All twelve fitted rankers have zero free-generation γ→δ Hit@5. For δ→γ, individual fitted values range from zero to 0.22% (zero to two of 900 queries), with no monotonic gain from increasing the ratio. The primary mixed/3 setting averages 0.11%; random/5, mixed/1 and mixed/5 each average 0.17%. Such small event differences do not establish a reproducible recovery advantage: paired intervals for every alternative negative setting versus the primary setting include zero in both directions and both fitted seeds.

Fixed-pool differences are larger but depend on the comparison. In random retrieval pools, random/3 gives two-seed mean ranker Hit@5 of 10.8% for γ→δ, compared with 8.2% under mixed/3. For δ→γ, random/5 gives 9.1%, compared with 7.1% under mixed/3. These descriptive rankings are not uniform across pools or seeds, and no setting supplies candidates absent from the frozen free-generation set. Negative-draw and fallback counts, selected steps and paired contrasts are retained in the numerical supplement. None of these constructed-negative comparisons identifies biological incompatibility.

Revision protocol

Nearest-input transfer uses the normalized full-CDR3 distance, lexicographic input ties and training-degree/lexicographic target ordering. One nearest input is primary; five is a fixed sensitivity. At most 64 unique training targets are returned. In seen-input/withheld-edge tasks the exact input is its own nearest neighbor, so one-neighbor transfer cannot recover its withheld edges. Primary component-disjoint transfer is also structurally zero because all test targets are absent from training. These zero outcomes do not identify a learned-model mechanism. The separate nearest-input partner-similarity retrieval score extends to the role-specific target universe using minimum target edit distance to the nearest input's known training partners. Neither neighborhood size nor distance rule was selected using test outcomes. Test memberships, selection repairs and the giant-component edge reservation are frozen in the revision split manifest. These are post-primary revision diagnostics.

Preserved original unseen-gamma diagnostic

The original selection partition contained one gamma query with 581 recorded partners and no ranking distractors. Its ranker selection MRR was identically one. Original conditional likelihood Hit@5 was 9/64 in each seed (14.06%), compared with 9/64 and 8/64 for constant source (14.06%/12.50%). These estimates are retained for traceability, not as robust unseen-gamma evidence. The repaired-selection refits preserve the original train, aggregate validation and test memberships and replace this diagnostic in the main text. Primary component-disjoint results remain unchanged. The revised selection has 31 gamma queries, 275 edges, a largest query with 137 partners and at least 118 eligible distractors per query. The tuning and calibration roles each contain 16 disjoint gamma queries and are not used for new tuning in this revision.

Interpretation of additional descriptive exports

Additional file 2 includes primary source and source-distance strata and a post-hoc source-balanced description. The latter averages nonempty source membership strata equally. Memberships can overlap, and very small source strata are unstable; this is neither a reweighted primary endpoint nor a source-independent inferential comparison. Post-hoc component-size tables group primary-test components into 1, 2-5, 6-20 and more than 20 edges, retaining the original query positives and saved candidates. Nonempty bins give counts and descriptive means without inferential intervals. The distant-edge recheck recovers 5, 5 and 3 primary edges, with no new claim of distant-sequence transfer. Within-giant-component estimates originate from one original component and deliberately have no inferential interval. Additional file 4 retains every revision query outcome, including zero events, all control conditions and paired contrasts. Additional file 3 records actual generator sequences and target tokens; ranker encoded-token counts were not logged and are not imputed.

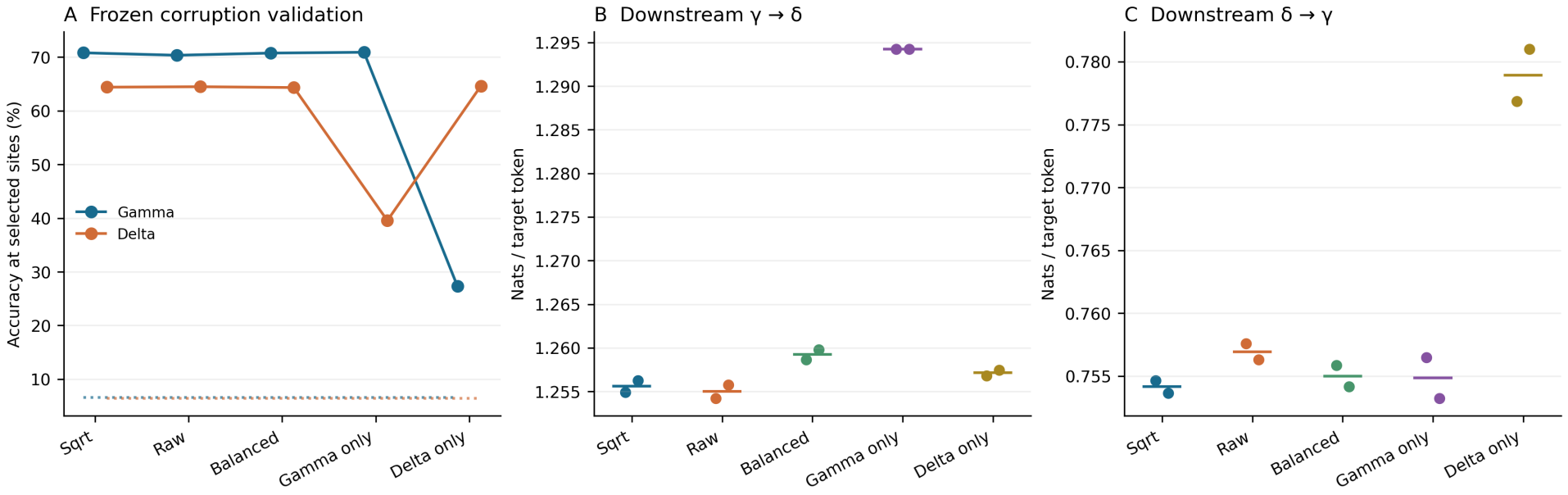


Figure S1. Pretraining mixture sensitivity. Dots retain fitted seeds. Equal sequence draws do not imply equal chain-specific or token exposure. Reconstruction is a single-chain endpoint.

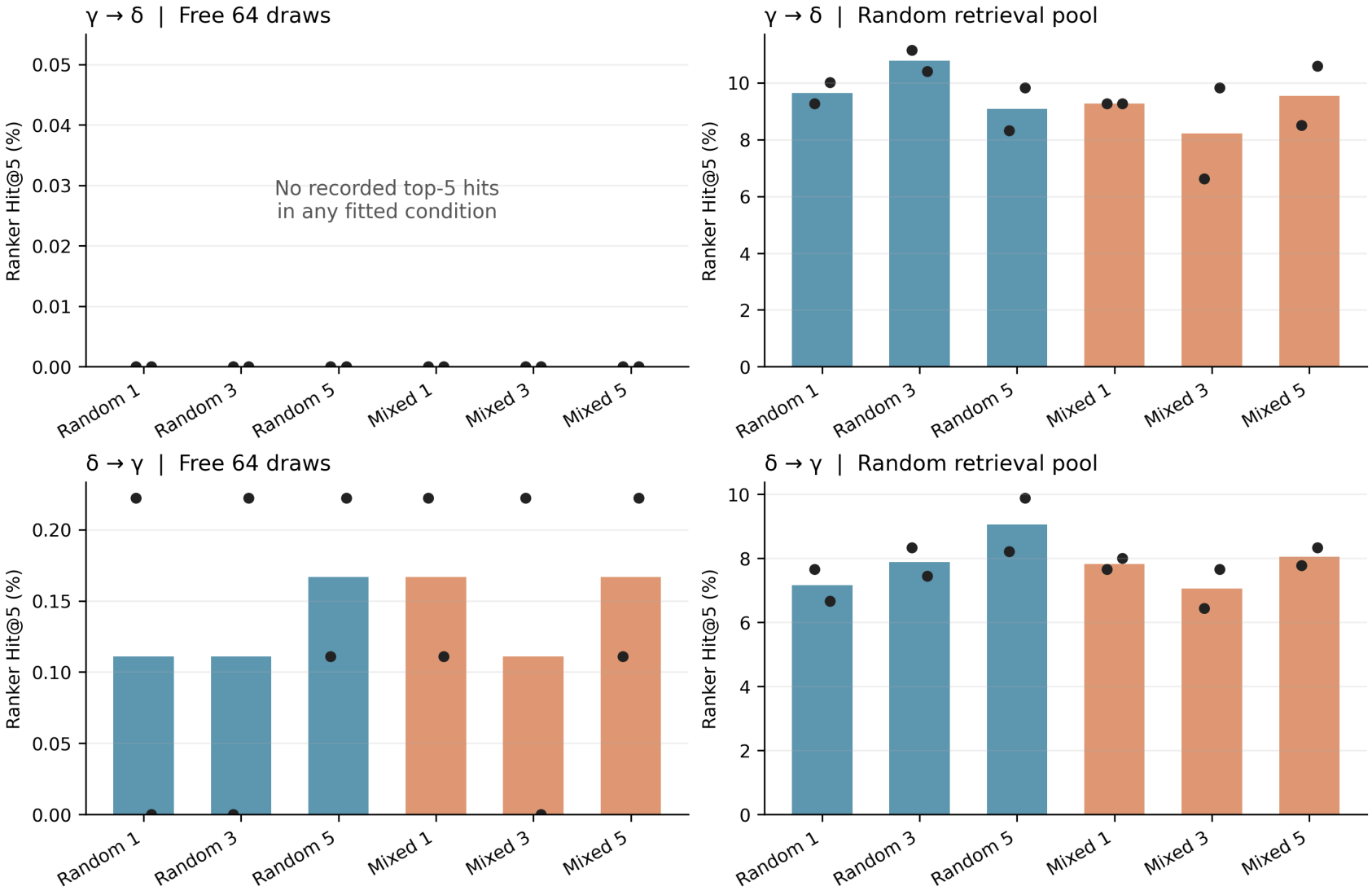


Figure S2. Ranker negative-sampling sensitivity. All settings reuse the same saved free candidates and frozen truth-containing retrieval pools. No ranker adds absent candidate sequences.